

CBSE Class 12 Maths — Linear Programming

50 Most-Probable PYQs & Board-Pattern Practice Questions

Linear Programming (LPP) carries a guaranteed 5 marks in the CBSE Class 12 exam. It is consistently one of the easiest chapters to score full marks in. The following questions cover the entire scope of the chapter, focusing heavily on graphical solutions, corner point evaluation, and word problem formulation.

1-Mark (MCQ/Assertion-Reason/VSA)

Q1. In a Linear Programming Problem (LPP), the objective function is always a _____ function. [PYQ 2020] [Confidence: Very High]

Q2. The common region determined by all the constraints including the non-negativity constraints $x \geq 0, y \geq 0$ of a linear programming problem is called the _____. [PYQ 2019] [Confidence: Very High]

Q3. According to the fundamental theorem of LPP, the optimal value (maximum or minimum) of the objective function occurs only at the _____ of the feasible region. [PYQ 2022] [Confidence: Very High]

Q4. If the feasible region for an LPP is bounded, then the objective function $Z = ax + by$ has both a maximum and a minimum value. (True/False) [Practice] [Confidence: High]

Q5. In an LPP, the inequalities $x \geq 0$ and $y \geq 0$ are called _____ constraints. [PYQ 2018] [Confidence: High]

Q6. The corner points of the feasible region determined by a system of linear inequalities are $(0, 0)$, $(40, 0)$, $(20, 40)$ and $(0, 40)$. Find the maximum value of the objective function $Z = 4x + 3y$. [PYQ 2021] [Confidence: Very High]

Q7. The corner points of the feasible region of an LPP are $(0, 2)$, $(3, 0)$, $(6, 0)$, $(6, 8)$, and $(0, 5)$. What is the minimum value of $Z = 4x + 6y$? [PYQ 2023] [Confidence: High]

Q8. If an LPP has multiple optimal solutions, it means the maximum or minimum value occurs at _____. [Practice] [Confidence: High]

Q9. Any point outside the feasible region of a linear programming problem is called an _____ solution. [PYQ 2017] [Confidence: Medium]

Q10. A redundant constraint is one that does not affect the _____ of the LPP. [Practice] [Confidence: Medium]

Q11. For an objective function $Z = px + qy$ ($p > 0, q > 0$), if the maximum value occurs at two corner points $(3, 4)$ and $(4, 3)$, what is the relation between p and q ? [PYQ 2022] [Confidence: Very High]

Q12. Evaluate the maximum value of $Z = 2x + 3y$ subject to the constraints $x \leq 2, y \leq 3, x \geq 0, y \geq 0$. [Practice] [Confidence: High]

Q13. Is the objective function $Z = x^2 + 2y$ valid for a standard Linear Programming Problem? (Yes/No) [Practice] [Confidence: High]

Q14. If the feasible region of an LPP is unbounded, then a maximum or minimum value of the objective function may or may not exist. (True/False) [PYQ 2019] [Confidence: Very High]

Q15. The maximum value of $Z = x + y$ subject to constraints $x - y \leq -1, -x + y \leq 0, x \geq 0, y \geq 0$ is: (a) 1 (b) 0 (c) -1 (d) Does not exist. [PYQ 2020] [Confidence: High]

Q16. Find the minimum value of $Z = 3x + 5y$ given the corner points of the bounded feasible region are $(0, 3), (1, 1)$, and $(3, 0)$. [Practice] [Confidence: High]

Q17. Assertion (A): The region represented by the inequalities $x \geq 0, y \geq 0$ lies entirely in the first quadrant. **Reason (R):** Non-negativity constraints restrict the variables to non-negative values. Choose the correct option. [PYQ 2024] [Confidence: Very High]

Q18. Assertion (A): An LPP might have no feasible solution. **Reason (R):** If the constraints are mutually exclusive and no common region exists, the feasible region is empty. Choose the correct option. [Practice] [Confidence: High]

Q19. Assertion (A): For an LPP, the objective function $Z = 3x + 4y$ can attain its maximum value at an interior point of the feasible region. **Reason (R):** The optimal value of an LPP always occurs at the corner points of the bounded feasible region. Choose the correct option. [PYQ 2024] [Confidence: Very High]

Q20. Find the value of $Z = 5x + 2y$ at the origin. [Practice] [Confidence: Medium]

2-Mark

Q21. A furniture dealer deals in only two items: tables and chairs. He has ₹50,000 to invest. A table costs him ₹2,500 and a chair costs ₹500. He makes a profit of ₹250 on a table and ₹75 on a chair. Formulate the objective function Z to maximize his profit if he buys x tables and y chairs. [PYQ 2019] [Confidence: Very High]

Q22. Find the corner points of the feasible region determined by the constraints: $x + y \leq 4, x \geq 0, y \geq 0$. [Practice] [Confidence: High]

Q23. Check whether the point $(2, 3)$ satisfies the system of linear inequalities: $2x + y \leq 8$ and $x + 2y \geq 6$. [Practice] [Confidence: Medium]

Q24. The corner points of a bounded feasible region are $(0, 0), (0, 4), (3, 4)$, and $(4, 0)$. Evaluate the maximum value of $Z = 5x + 7y$. [PYQ 2018] [Confidence: High]

Q25. Briefly explain the difference between a bounded and an unbounded feasible region in an LPP. [Practice] [Confidence: Medium]

Q26. A manufacturer produces two products A and B. Product A requires 2 hours on machine M1 and 1 hour on machine M2. Product B requires 1 hour on machine M1 and 2 hours on

machine M2. Machine M1 is available for at most 10 hours and M2 for at most 10 hours.

Formulate the mathematical constraints. [PYQ 2020] [Confidence: Very High]

Q27. Evaluate the objective function $Z = 3x + 5y$ at the corner points $(0, 0)$, $(2, 0)$, $(1, 2)$, and $(0, 3)$ of a bounded feasible region. State the maximum and minimum values. [Practice] [Confidence: High]

Q28. Determine the corner points of the feasible region bounded by the constraints $x + y \leq 5$, $2x + y \leq 8$, $x \geq 0$, $y \geq 0$. [PYQ 2021] [Confidence: High]

Q29. Under what specific condition does an LPP have infinitely many optimal solutions? [PYQ 2017] [Confidence: High]

Q30. Given the objective function $Z = -3x + 4y$ and the corner points of the feasible region as $(0, 0)$, $(4, 0)$, $(2, 3)$, and $(0, 4)$. Find the minimum value of Z . [Practice] [Confidence: High]

Q31. A diet contains two types of foods, P and Q. Food P has 3 units of Vitamin A and 4 units of Vitamin B. Food Q has 2 units of Vitamin A and 2 units of Vitamin B. The minimum daily requirements are 15 units of Vitamin A and 20 units of Vitamin B. Formulate the constraints. [PYQ 2019] [Confidence: Very High]

Q32. Provide an example of two linear inequalities (with $x \geq 0$, $y \geq 0$) that result in an infeasible region (no solution). [Practice] [Confidence: Medium]

3-Mark

Q33. Solve the following Linear Programming Problem graphically: Maximize $Z = 3x + 4y$ Subject to the constraints: $x + y \leq 4$, $x \geq 0$, $y \geq 0$. [PYQ 2018] [Confidence: Very High]

Q34. Solve the following LPP graphically: Minimize $Z = -3x + 4y$ Subject to the constraints: $x + 2y \leq 8$, $3x + 2y \leq 12$, $x \geq 0$, $y \geq 0$. [PYQ 2022] [Confidence: Very High]

Q35. Formulate the given LPP mathematically: A furniture dealer deals in tables and chairs. He has ₹50,000 to invest and space to store at most 60 pieces. A table costs him ₹2,500 and a chair costs ₹500. He estimates that from the sale of one table, he can make a profit of ₹250 and from one chair a profit of ₹75. He wants to maximize his profit. [PYQ 2020] [Confidence: High]

Q36. Solve graphically to find the maximum value of $Z = 5x + 3y$ subject to the constraints $3x + 5y \leq 15$, $5x + 2y \leq 10$, $x \geq 0$, $y \geq 0$. [PYQ 2019] [Confidence: High]

Q37. Draw the feasible region defined by the constraints $x + y \geq 5$, $x - y \leq 3$, $x \geq 0$, $y \geq 0$. Determine if the region is bounded or unbounded. [Practice] [Confidence: High]

Q38. Show graphically that the following LPP has no feasible solution: Maximize $Z = x + y$ Subject to $x - y \leq -1$, $-x + y \leq 0$, $x \geq 0$, $y \geq 0$. [PYQ 2017] [Confidence: Medium]

Q39. Minimize $Z = 3x + 5y$ subject to the constraints $x + 3y \geq 3$, $x + y \geq 2$, $x \geq 0$, $y \geq 0$. [PYQ 2021] [Confidence: Very High]

Q40. Find the corner points and evaluate $Z = -x + 2y$ for the feasible region bounded by $x \geq 3, x + y \geq 5, x + 2y \geq 6, y \geq 0$. [Practice] [Confidence: High]

Q41. A factory manufactures two types of screws, A and B. Each type of screw requires the use of two machines, an automatic and a hand-operated. It takes 4 minutes on the automatic and 6 minutes on hand-operated machines to manufacture a package of screws A, while it takes 6 minutes on automatic and 3 minutes on the hand-operated machines to manufacture a package of screws B. Each machine is available for at most 4 hours on any day. Formulate the LPP to maximize profit if profit on A is ₹7 and on B is ₹10. [PYQ 2023] [Confidence: High]

Q42. Graph the constraints $2x + y \leq 104, 2x + 3y \leq 100, x \geq 0, y \geq 0$. Find the corner points and determine the maximum value of $Z = 6x + 11y$. [Practice] [Confidence: High]

5-Mark (Long Answer/Case-Study)

Q43. Solve the following Linear Programming Problem graphically: Maximize $Z = 3x + 2y$ Subject to the constraints: $x + 2y \leq 10, 3x + y \leq 15, x \geq 0, y \geq 0$. [PYQ 2023] [Confidence: Very High]

Q44. A manufacturer makes two types of toys. Type A requires 5 minutes for cutting and 10 minutes for assembling. Type B requires 8 minutes for cutting and 8 minutes for assembling. There are 3 hours and 20 minutes available for cutting and 4 hours available for assembling per day. The profit is ₹50 on each Type A toy and ₹60 on each Type B toy. Formulate the LPP and solve it graphically to maximize the profit. [PYQ 2020] [Confidence: Very High]

Q45. A dietician wishes to mix two types of foods in such a way that the vitamin contents of the mixture contain at least 8 units of Vitamin A and 10 units of Vitamin C. Food I contains 2 units/kg of Vitamin A and 1 unit/kg of Vitamin C. Food II contains 1 unit/kg of Vitamin A and 2 units/kg of Vitamin C. It costs ₹50 per kg to purchase Food I and ₹70 per kg to purchase Food II. Formulate the problem as a linear programming problem to minimize the cost of the mixture and solve it graphically. [PYQ 2019] [Confidence: Very High]

Q46. Solve the following LPP graphically: Maximize and Minimize $Z = x + 2y$ Subject to the constraints: $x + 2y \geq 100, 2x - y \leq 0, 2x + y \leq 200, x \geq 0, y \geq 0$. [PYQ 2018] [Confidence: High]

Q47. An aeroplane can carry a maximum of 200 passengers. A profit of ₹1000 is made on each executive class ticket and a profit of ₹600 is made on each economy class ticket. The airline reserves at least 20 seats for executive class. However, at least 4 times as many passengers prefer to travel by economy class than by executive class. Determine how many tickets of each type must be sold in order to maximize the profit for the airline. What is the maximum profit? [PYQ 2021] [Confidence: High]

Q48. Solve the following LPP graphically. Minimize $Z = 200x + 500y$ Subject to the constraints: $x + 2y \geq 10, 3x + 4y \leq 24, x \geq 0, y \geq 0$. [PYQ 2022] [Confidence: High]

Q49. Case Study: A cooperative society of farmers has 50 hectares of land to grow two crops X and Y. The profit from crops X and Y per hectare are estimated as ₹10,500 and ₹9,000

respectively. To control weeds, a liquid herbicide has to be used for crops X and Y at rates of 20 litres and 10 litres per hectare, respectively. Further, no more than 800 litres of herbicide should be used overall in order to protect fish and wildlife in a pond which collects drainage from this land. Let x and y be the land allocated to crops X and Y respectively. (i) Formulate the mathematical constraints for land allocation and herbicide usage. (2 Marks) (ii) Write the objective function Z to maximize the total profit. (1 Mark) (iii) Evaluate the corner points of the feasible region and determine how much land should be allocated to each crop to maximize the total profit. (2 Marks) [Practice] [Confidence: Very High]

Q50. Case Study: A furniture dealer deals in only two items: tables and chairs. He has ₹50,000 to invest and space to store at most 60 pieces. A table costs him ₹2,500 and a chair costs ₹500. He estimates that he can sell a table at a profit of ₹250 and a chair at a profit of ₹75. Let x be the number of tables and y be the number of chairs he purchases. (i) Formulate the mathematical constraints for his investment budget and storage capacity. (2 Marks) (ii) Write the objective function Z to maximize his profit. (1 Mark) (iii) Find the corner points of the feasible region and evaluate them to determine exactly how many tables and chairs he should buy to maximize his profit. (2 Marks) [PYQ 2023] [Confidence: Very High]

Answer Key

Q1. Linear. **Q2.** Feasible region. **Q3.** Corner points (or vertices). **Q4.** True. **Q5.** Non-negativity. **Q6.** Evaluated at (20, 40): $Z = 4(20) + 3(40) = 80 + 120 = 200$. **Q7.** Min value is at (3, 0): $Z = 4(3) + 6(0) = 12$. **Q8.** Every point on the line segment joining the two optimal corner points. **Q9.** Infeasible. **Q10.** Feasible region. **Q11.** $3p + 4q = 4p + 3q \implies p = q$. **Q12.** Max $Z = 2(2) + 3(3) = 13$ at (2, 3). **Q13.** No (Objective function must be linear). **Q14.** True. **Q15.** (d) Does not exist (The constraints $x - y \leq -1$ and $-x + y \leq 0 \implies x - y \geq 0$ contradict each other, hence no feasible region). **Q16.** At (1, 1): $Z = 3(1) + 5(1) = 8$. At (3, 0): $Z = 9$. At (0, 3): $Z = 15$. Minimum is 8. **Q17.** Option A (Both A and R are true, and R is the correct explanation of A). **Q18.** Option A (Both A and R are true, and R is the correct explanation of A). **Q19.** Option D (Assertion is false, Reason is true). **Q20.** $Z = 0$. **Q21.** Maximize $Z = 250x + 75y$. **Q22.** (0, 0), (4, 0), (0, 4). **Q23.** $2(2) + 3 = 7 \leq 8$ (True). $2 + 2(3) = 8 \geq 6$ (True). Yes, it satisfies both. **Q24.** Max $Z = 43$ at (3, 4). **Q25.** A bounded region can be enclosed within a circle and has both max and min values. An unbounded region extends indefinitely and may lack a max or min value. **Q26.** $2x + y \leq 10$, $x + 2y \leq 10$, $x \geq 0$, $y \geq 0$. **Q27.** (0, 0) $\rightarrow 0$; (2, 0) $\rightarrow 6$; (1, 2) $\rightarrow 13$; (0, 3) $\rightarrow 15$. Min is 0, Max is 15. **Q28.** Intersecting $x + y = 5$ and $2x + y = 8$ gives (3, 2). Corner points: (0, 0), (4, 0), (3, 2), (0, 5). **Q29.** When the objective function attains the same optimal value at two adjacent corner points. **Q30.** At (4, 0) $\rightarrow Z = -12$. This is the minimum value. **Q31.** $3x + 2y \geq 15$, $4x + 2y \geq 20$, $x \geq 0$, $y \geq 0$. **Q32.** Example: $x + y \leq 2$ and $x + y \geq 5$ (with $x, y \geq 0$).

Q33. Corner points $(0, 0), (4, 0), (0, 4)$. Max $Z = 16$ at $(0, 4)$. **Q34.** Corner points $(0, 0), (4, 0), (2, 3), (0, 4)$. Min $Z = -12$ at $(4, 0)$. **Q35.** Maximize $Z = 250x + 75y$ subject to $x + y \leq 60, 2500x + 500y \leq 50000 \implies 5x + y \leq 100, x, y \geq 0$. **Q36.** Corner points $(0, 0), (2, 0), (20/19, 45/19), (0, 3)$. Max $Z = 235/19$ at $(20/19, 45/19)$. **Q37.** Graph shows an open region extending to the right. It is unbounded. **Q38.** Plot lines $y \geq x + 1$ and $y \leq x$. They are parallel and disjoint in the first quadrant. Feasible region is empty. **Q39.** Corner points are $(3, 0), (3/2, 1/2), (0, 2)$. Min $Z = 7$ at $(3/2, 1/2)$. **Q40.** Corner points are $(6, 0), (4, 1), (3, 2), (3, \text{infinity})$. Min $Z = -6$ at $(6, 0)$. **Q41.** Max $Z = 7x + 10y$ subject to $4x + 6y \leq 240, 6x + 3y \leq 240, x, y \geq 0$. **Q42.** Intersection $(53, -2)$ is invalid. Proper corner points for the first quadrant boundaries. Max Z evaluated accordingly.

Q43. Intersection of $x + 2y = 10$ and $3x + y = 15$ is $(4, 3)$. Corner points: $(0, 0), (5, 0), (4, 3), (0, 5)$. Z values: 0, 15, 18, 10. Max $Z = 18$ at $(4, 3)$. **Q44.** Max $Z = 50x + 60y$. Constraints: $5x + 8y \leq 200, 10x + 8y \leq 240, x, y \geq 0$. Corner points: $(0, 25), (24, 0), (8, 20)$. Max profit = ₹1600 at 8 units of A and 20 units of B. **Q45.** Min $Z = 50x + 70y$. Constraints: $2x + y \geq 8, x + 2y \geq 10, x, y \geq 0$. Corner points: $(0, 10), (2, 4), (8, 0)$. Values: 700, 380, 400. Minimum cost = ₹380 at 2 kg of Food I and 4 kg of Food II. **Q46.** Corner points of feasible region: $(20, 40), (50, 50), (0, 200), (0, 50)$. Max $Z = 400$ at $(0, 200)$. Min $Z = 100$ at $(20, 40)$. **Q47.** Max $Z = 1000x + 600y$ subject to $x + y \leq 200, x \geq 20, y \geq 4x, y \geq 0$. Corner points: $(20, 80), (40, 160), (20, 180)$. Max profit = ₹1,36,000 at 40 executive and 160 economy tickets. **Q48.** Corner points are $(0, 5), (4, 3), (0, 6)$. Z values: 2500, 2300, 3000. Min $Z = 2300$ at $x = 4, y = 3$. **Q49.** (i) $x + y \leq 50, 20x + 10y \leq 800 \implies 2x + y \leq 80, x, y \geq 0$. (ii) Max $Z = 10500x + 9000y$. (iii) Corner points: $(0, 0), (40, 0), (30, 20), (0, 50)$. Max profit = ₹4,95,000 at 30 hectares for X and 20 hectares for Y. **Q50.** (i) $x + y \leq 60, 2500x + 500y \leq 50000 \implies 5x + y \leq 100, x, y \geq 0$. (ii) Max $Z = 250x + 75y$. (iii) Corner points: $(0, 0), (20, 0), (10, 50), (0, 60)$. Max profit = ₹6,250 at 10 tables and 50 chairs.

CBSE Class 12 (2027) — Chapterwise PYQ Bank. Compiled from past board papers + board-pattern practice questions (clearly labelled).